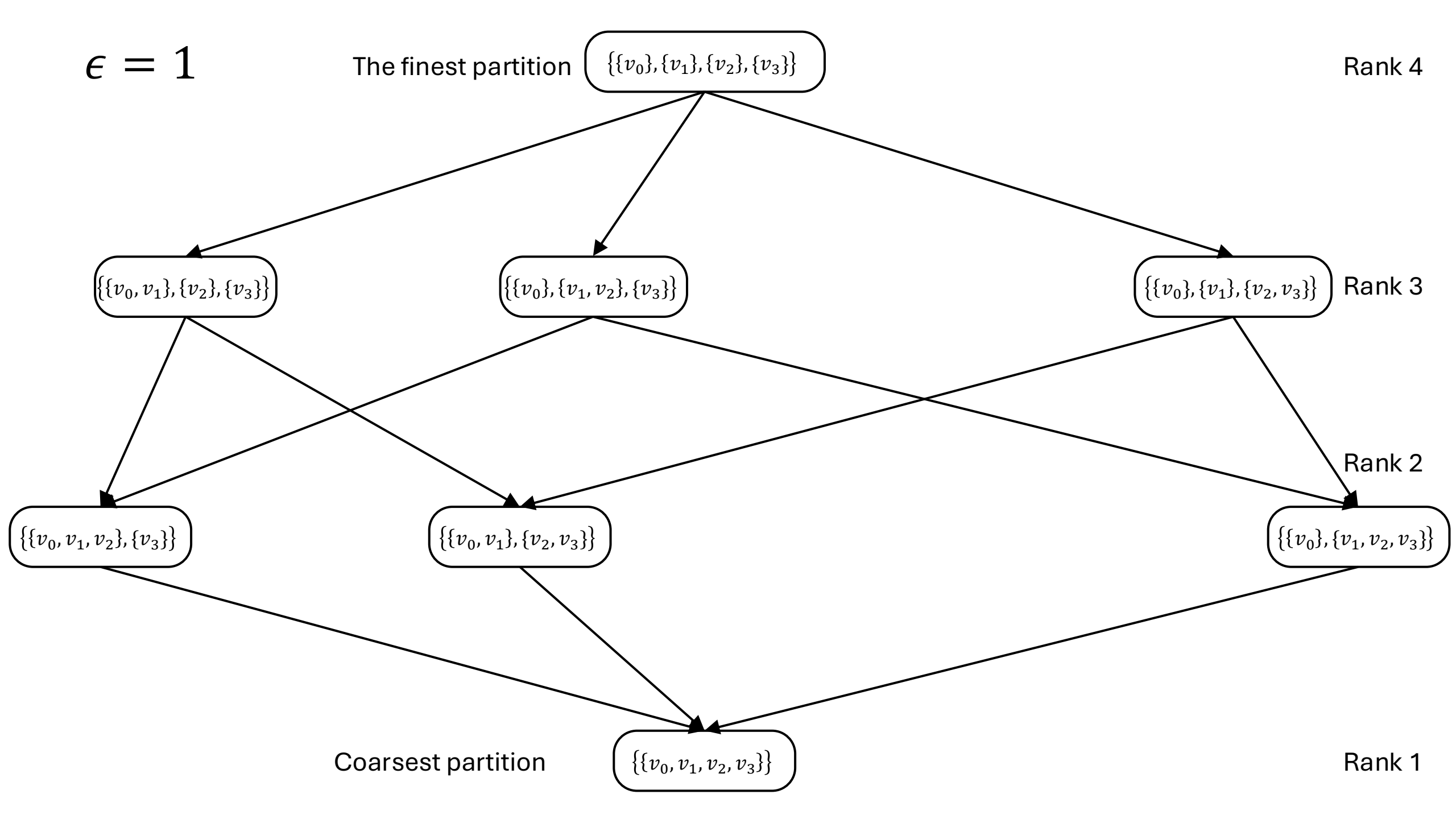

$\epsilon = 1$
The finest partition
$\{\{v_0\},\{v_1\},\{v_2\},\{v_3\}\}$
Rank 4
$\{\{v_0,v_1\},\{v_2\},\{v_3\}\}$
$\{\{v_0\},\{v_1,v_2\},\{v_3\}\}$
$\{\{v_0\},\{v_1\},\{v_2,v_3\}\}$
Rank 3
Rank 2
$\{\{v_0,v_1,v_2\},\{v_3\}\}$
$\{\{v_0,v_1\},\{v_2,v_3\}\}$
$\{\{v_0\},\{v_1,v_2,v_3\}\}$
Coarsest partition
$\{\{v_0,v_1,v_2,v_3\}\}$
Rank 1